AEGNER BILLIK

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Summary

Informatics Engineering graduate from Muhammadiyah University of Bandung (GPA: 3.75/4.00, thesis and coursework completed, graduating January 2027) with full-time availability. Proven hands-on experience in fullstack web architecture, distributed database systems, and hardware-software integration. Track record of engineering real-world production platforms, including an offline-capable enterprise ERP & POS system with native Bluetooth thermal printing, multi-tenant academic CMS portals, and industrial IoT telemetry dashboards. Bangkit Academy 2024 Machine Learning Graduate (Google, GoTo, Traveloka) and National P2MW Grant Recipient. Dedicated to writing clean, modular code that solves core business challenges.

Education

Muhammadiyah University of Bandung *Sep 2022 – Present (Graduating Jan 2027)*

Bachelor's Degree in Informatics Engineering (Computer Science) | GPA : 3.75 / 4.00

- Relevant Coursework: Software Engineering, Database Systems, Object-Oriented Programming, Web Architecture, Applied Machine Learning, Distributed Systems.

Professional Experience

Koperasi Universitas Muhammadiyah Bandung | Fullstack Software Engineer *Oct 2025 – Present*

- Architected and built an end-to-end enterprise management system handling member lifecycle, savings accounts, loan processing, automated payroll deductions, and General Ledger accounting.
- Developed an offline-first Point of Sale (POS) tablet application leveraging IndexedDB for local transaction caching and automated conflict-free synchronization upon network recovery.
- Implemented native hardware communication with Bluetooth thermal receipt printers via Java RFCOMM SPP socket protocols.
- Engineered relational database schemas in PostgreSQL with transaction integrity to support multi-batch supplier consignment, inventory auditing, and annual SHU RAT profit distributions.

Universitas Muhammadiyah Bandung | Web Developer (Contract) *Jun 2026 – Jul 2026*

- Developed a dynamic academic web portal and content management system (CMS) for university study programs (Biotechnology, Fashion Craft, and FST).
- Leveraged Next.js 14 App Router, TypeScript, and Tailwind CSS to achieve fast page loads, responsive layouts, and streamlined document archival for accreditation readiness.
- Integrated the portal with internal faculty databases to synchronize faculty research publications and curriculum metrics.

Web Design Laboratory | Laboratory Assistant *Mar 2024 – Feb 2025*

- Mentored 40+ undergraduate students across two semesters in modern web development fundamentals (HTML5, CSS3 Grid/Flexbox, responsive design, and JavaScript DOM manipulation).
- Designed structured weekly grading rubrics, evaluated student laboratory module submissions, and conducted hands-on code debugging sessions.

Bangkit Academy led by Google, Tokopedia, GoTo, & Traveloka | Machine Learning Cohort *Aug 2024 – Dec 2024*

- Completed 900+ curriculum hours covering supervised learning, deep neural networks, computer vision, natural language processing, and model deployment.
- Built and deployed end-to-end machine learning pipelines using TensorFlow, Python, and Google Cloud Platform.

Projects

Real-Time Factory QC Monitoring System | IoT & Fullstack Engineer *May 2024 – Aug 2024*

- Designed an industrial quality control tracking system integrating ESP32 microcontrollers, RFID RC522 card readers, and a centralized web dashboard.
- Implemented bidirectional WebSockets telemetry using Django Channels and Redis message broker, enabling live sub-second updates of assembly line throughput without page reloads.

- Programmed firmware in C/C++ with hardware debouncing to capture technician shift logs and quality inspection timestamps.

OCR e-KTP Identity Extraction System (Final Thesis) | Lead Developer *2024 – 2026*

- Engineered an automated optical character recognition pipeline to digitize national identity cards (e-KTP) for the university new student admissions (PMB) portal.
- Applied computer vision pre-processing filters (noise reduction, adaptive thresholding, deskewing) with OpenCV to optimize text clarity on smartphone captures.
- Constructed a RegEx parsing engine to reliably extract structured 16-digit NIK, full name, and demographic fields, minimizing manual data-entry errors by 80%.

Audio Music Genre Classification | Machine Learning Engineer *2024*

- Developed an audio classification pipeline using PyTorch and Librosa to classify raw audio tracks across 10 musical genres on the GTZAN dataset.
- Extracted Mel-Frequency Cepstral Coefficients (MFCC) spectrograms and trained a Convolutional Neural Network (CNN) with dropout regularization.
- Built an interactive web interface using Streamlit featuring 3-second windowing and majority voting for robust full-track song inference.

Honors & Awards

- **National P2MW Grant Recipient (2024):** Awarded national business innovation funding by the Ministry of Education, Culture, Research, and Technology (Kemendikbudristek RI) for the digital startup platform "Forum Kampus".
- **National Finalist Delegate – KMI Expo XV (2024):** Selected as official university delegate and national finalist at the 15th Indonesian Student Entrepreneurship (KMI) Expo held at Halu Oleo University, Kendari.

Skills & Certifications

Programming Languages: JavaScript (ES6+), TypeScript, PHP, Python, SQL, C/C++ (ESP32), Java (Android basics).

Frameworks & Web Tech: React.js, Next.js 14, Laravel 11, Node.js, Express, Tailwind CSS, Vite, RESTful APIs, WebSockets.

Databases & Storage: PostgreSQL, MySQL, Supabase, Redis, IndexedDB (Offline caching).

AI, Vision & Embedded: OpenCV, Tesseract OCR, PyTorch, TensorFlow, Microcontroller ESP32, RFID.

Developer Tools: Git/GitHub, Postman, Linux/CLI, Docker (basics), CI/CD pipelines.

Certifications: Google Bangkit ML Graduate, Google IT Automation with Python, DeepLearning.AI ML Specialization, Jonas Schmedtmann Node.js Bootcamp, Stephen Grider SQL & PostgreSQL.

Languages: Indonesian (Native), English (Professional Working Proficiency).